DLP for WhatsApp

1.0.1

Product Guide/User Guide

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Revision A

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Chapter 1: Introduction

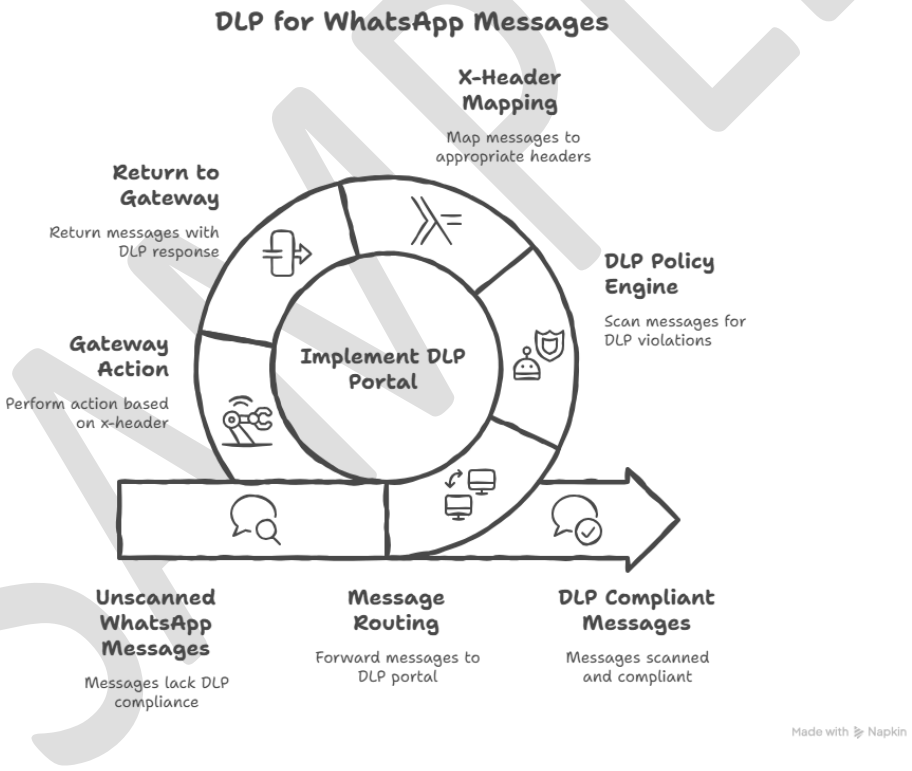
DLP for WhatsApp helps prevent data leaks via the WhatsApp messaging channel.

It works with outbound WhatsApp messages sent from organizational devices or business accounts, scanning them against central DLP policies before delivery. The solution integrates with a DLP portal running on AWS, where messages are evaluated and a recommended action is returned to the organization's WhatsApp gateway in a custom header (X-DLP-WhatsApp).

The organization then decides the actual action to take on each message based on its own policies. The action determined by the scan (allow, block, quarantine, encrypt) is not enforced automatically; it is provided only as a recommendation.

Conceptual Architecture

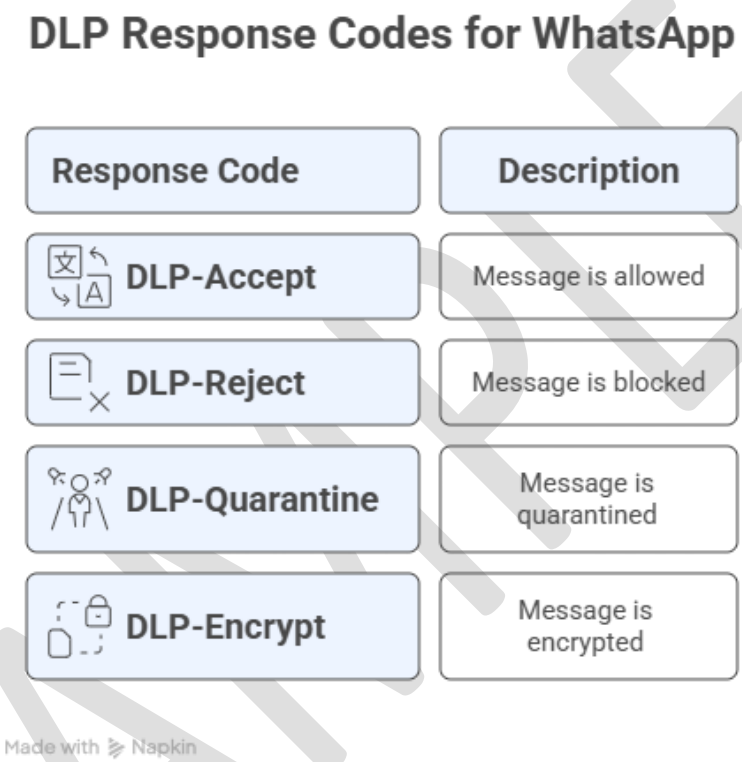
DLP for WhatsApp integrates with the DLP portal and the organization's WhatsApp gateway to protect outbound WhatsApp messages from sensitive data leaks.



1. A user initiates a WhatsApp message from a mobile device, web client, or desktop WhatsApp application.
2. In the WhatsApp gateway environment, message routing rules are configured to forward and receive outbound WhatsApp messages to and from DLP for WhatsApp (AWS-hosted DLP portal).
3. When an outbound WhatsApp message enters the organization's WhatsApp gateway, it is routed to the DLP portal according to these message routing rules.
4. In the DLP portal, outbound WhatsApp messages are scanned by the DLP Policy Engine.

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5. Based on the tenant-configured DLP policies, outbound WhatsApp messages are mapped to the appropriate X headers.
6. After the DLP scan is complete, DLP for WhatsApp returns the messages to the organization's WhatsApp gateway with the corresponding X header.
7. The organization's WhatsApp gateway receives the DLP-processed messages, reads the custom header containing the DLP response, and performs the appropriate action based on that X header.



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Features and Benefits

- **Protect WhatsApp from data loss**
  WhatsApp is a widely used channel for both business and personal communication, which makes it vulnerable to accidental or deliberate data leaks. DLP for WhatsApp applies targeted policies based on sender and recipient attributes, reducing the risk that sensitive information is shared inappropriately.

- **Agentless and hassle-free data leak prevention**
  DLP for WhatsApp does not require any agent installation on user devices, simplifying deployment and maintenance. It supports all WhatsApp clients and ensures that consistent DLP policies are enforced across mobile, web, and desktop environments.

- **Flexible and customizable actions**
  Administrators can choose from different actions such as reject, quarantine, or encrypt, depending on business and compliance requirements. Policies can also be configured to avoid quarantining messages when data sovereignty or storage restrictions apply.

- **Upcoming support for additional X-header actions**
  DLP for WhatsApp is designed to support additional X-header actions such as drop attachment and quarantine or reject with user notification. These enhancements will give organizations more granular control over how sensitive business communications are handled and protected against data loss.

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Chapter 2: Product Updates

This section details the new and updated features, known and resolved issues.

What's new

- **Multi-platform WhatsApp protection**
  DLP for WhatsApp now offers multi-platform support, extending chat security to both personal and business WhatsApp communication across mobile, web, and desktop clients.
- **X-header support for dropping sensitive attachments**
  DLP for WhatsApp now supports an X-header action to drop sensitive attachments while notifying the user, helping prevent data loss through high-risk files.
- **Granular content classifiers for WhatsApp messages**
  Added support for advanced data classifiers (PII, payment data, health data, source code) tuned for short chat messages and emojis, improving detection quality in WhatsApp conversations.
- **Context-aware policy engine for group chats**
  Policies can now distinguish between 1:1 chats, internal groups, and external customer groups, allowing different actions (for example, allow internal groups but quarantine messages to external groups containing confidential terms).
- **Enhanced AWS auditing and logging**
  Auditing capabilities for the AWS-hosted DLP portal have been enhanced to support more comprehensive security and compliance reporting.
  You can now view detailed log data about each scanned message, including timestamps and X-header mapping information.

Resolved issues

- **User notifications for rejected or quarantined messages**
  Users are now notified when their WhatsApp message is rejected or quarantined based on DLP policy, improving transparency and reducing help desk tickets.
- **False positives on short numeric messages**
  Resolved an issue where short numeric chats (such as OTPs or reference IDs) were frequently classified as payment card data, leading to unnecessary encryption or quarantine.
- **Delay when scanning large media files**
  Optimized media inspection so that large video and image attachments are scanned in a streaming fashion, significantly reducing end-user latency for outbound messages.
- **Missing audit entries for blocked messages**
  Corrected a logging defect where some DLP-Reject actions were enforced at the gateway but did not appear in the centralized incident report, improving audit completeness.

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Known issues

- **End-to-end encryption constraints**
  Due to WhatsApp's end-to-end encryption, content inspection is only available for traffic that passes through the integrated gateway or capture layer; messages sent outside this path cannot be scanned.
- **Limited visibility into unmanaged devices**
  Messages sent from unmanaged or personal devices that do not use the corporate WhatsApp gateway may bypass DLP controls unless combined with MDM or additional network controls.
- **Partial support for third-party WhatsApp clients and automation tools**
  Some unofficial clients, bots, or automation frameworks may not preserve the X-DLP-WhatsApp header, which can prevent downstream enforcement on the gateway.
- **High-volume group chats may generate alert noise**
  In large broadcast or campaign groups, a single misconfigured policy can generate many low-value incidents; tuning thresholds and using metadata-only policies is recommended.

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Chapter 3: Getting Started

This section describes the system requirements and additional information needed to start using DLP for WhatsApp.

System requirements

To enable DLP for WhatsApp, you need the following:

- A supported DLP platform with WhatsApp integration enabled and licensed.
- An organization-managed WhatsApp gateway or WhatsApp Business API environment through which outbound WhatsApp messages are routed.
- A valid cloud infrastructure account (for example, an AWS account) to host the DLP portal and policy engine, with secure connectivity from the organization's WhatsApp gateway.
- Network connectivity that allows outbound traffic from the organization's WhatsApp gateway to the DLP portal over the required ports and protocols.
- A modern web browser to access the DLP for WhatsApp administration portal, such as:
    - Google Chrome
    - Mozilla Firefox
    - Microsoft Edge
- For optimal console usability, a minimum screen resolution of 1360 × 768 is recommended, with 1920 × 1080 or higher preferred.

Signing in to the DLP Portal

To sign in to the cloud-hosted DLP portal, which is the common management interface for all DLP for WhatsApp services.

Note:

- ***If you are an existing* DLP *customer and your environment has already been migrated to the new DLP portal, you can log in directly and navigate to the DLP for WhatsApp service using the steps below (see steps 6 and 7).***
- ***If you are a new DLP user, follow steps 1 to 7 to activate your account, sign in to the portal, and open the DLP for WhatsApp service.***

1. When your administrator creates your account, you receive a welcome email with login credentials.
2. Click the **Accept invitation** link in the email to access the portal and sign in for the first time.
3. On the account activation page, enter your credentials, then click **Access account**.
4. You are prompted to set a new password that meets the following conditions:

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- Minimum 8 characters and maximum 20 characters
- At least one uppercase and one lowercase letter
- At least one digit
- At least one special character from the following set: ! @ # $ % & ? = [ ] < > { }

5. Enter your new password and click **Confirm**.
6. After successful sign-in, you are redirected to the main landing page of the DLP portal.
7. In the left navigation menu, expand the available services and click **DLP for WhatsApp**.

This opens the main **DLP for WhatsApp** page, where you can begin configuring policies, gateways, and incident handling settings for WhatsApp traffic.

Checking tenant information

Use the DLP portal to verify that your tenant and licenses are correctly provisioned for WhatsApp protection.

Steps

1. Sign in to the DLP portal with your administrator credentials.
2. In the DLP portal, navigate to **User > Tenant information**.
3. Verify that **DLP for WhatsApp** is listed as an active service for your tenant.
4. If your tenant or license information is not updated, appears incorrect, or has expired, contact the technical support team.

(Optional) Notification settings for DLP incidents

You can configure notifications so that administrators and/or users are informed when a WhatsApp message violates a DLP policy.

1. Configure the notification service that the DLP platform will use to send alerts (for example, email, chat, or ticketing integration). For more information, see *Configure the notification service*.
2. Define the notification templates that will be sent when a DLP incident is generated, including details such as action taken (allow, reject, quarantine, encrypt), policy name, and timestamp. For more information, see *Define the notification templates*.
3. Optionally configure user-facing notifications so that senders are informed when their WhatsApp messages are rejected, quarantined, or modified due to DLP policy enforcement. For more information, see *Configure user-facing notifications*.

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Chapter 4: Configuring DLP for WhatsApp

This chapter describes how to configure DLP for WhatsApp in your organization's environment using a WhatsApp gateway and a cloud-hosted DLP portal.

Configuring WhatsApp routing in the DLP portal

To enable DLP for WhatsApp, configure message routing settings in the DLP portal so that outbound WhatsApp traffic is sent to the DLP Policy Engine and then returned to your gateway.

Steps

1. Sign in to the **DLP portal** with your administrator credentials.
2. In the navigation menu, browse to **Messaging > WhatsApp**.
3. In the **Customer connection** section, select **WhatsApp gateway** as the source platform.
4. In the **Domains/identifiers** section, enter the WhatsApp Business numbers, API identifiers, or tenant IDs to which DLP policies should apply.
5. (Optional) In the **Return path** section, specify the IP address or DNS name and port of the organization's WhatsApp gateway or downstream messaging gateway to which processed messages will be returned.
6. Click **Save**.

DLP for WhatsApp is now enabled at the portal level for the specified gateway endpoints.

Configuring outbound routing on the WhatsApp gateway

The outbound routing configuration ensures that WhatsApp messages are sent from the organization's WhatsApp gateway to the DLP portal for inspection before they are delivered to external recipients.

Steps

1. In the WhatsApp management console, browse to S**ettings > Routing > Outbound rule.**
2. Click **Add rule** and then enter a descriptive name.
3. In the **Conditions** section, configure the rule to match:
    1. **Direction**: Outbound.
    2. **Recipients**: Select the DLP portal connection.
4. In the **Next hop** section, choose **Route to external service/proxy**.
5. Enter the **DNS name** or **IP address** and **port** of the DLP portal endpoint (for example, dlp-whatsapp.company.com:443).
6. Enable **TLS/HTTPS** to securely transmit messages between the gateway and the DLP portal.

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7. Set reasonable timeouts for connecting to the DLP portal and retry counts.
8. Click **Save**.
9. Then change the rule status to **Enabled** to start processing outbound traffic.

Configuring inbound processing on the WhatsApp gateway

After inspection, the DLP portal returns messages to the organization's WhatsApp gateway with an X-DLP-WhatsApp header that contains the recommended action (for example, accept, reject, quarantine, encrypt). The gateway must be configured to read this header and apply the correct handling.

Steps

1. In the WhatsApp management console, browse to **Settings > Routing > Inbound rule.**
2. Click **Add rule**. then enter a descriptive name.
3. In the **Conditions** section, configure the rule to match:
    - Set Direction to **Inbound**.
    - In **Source**, specify the DLP portal IP / DNS so only DLP-processed messages are matched.
4. In the **Condition** section, select **Message header equals/contains** and specify header name **X-DLP-WhatsApp**.
5. Create actions based on the value of the header:
    - If X-DLP-WhatsApp = **DLP-Accept** → Deliver the message normally.
    - If X-DLP-WhatsApp = **DLP-Reject** → Block the message and optionally notify the sender.
    - If X-DLP-WhatsApp = **DLP-Quarantine** → Move the message to a Quarantine queue for security review.
    - If X-DLP-WhatsApp = **DLP-Encrypt** → Apply encryption before sending the message externally.
6. Set the rule **High** in the processing order.
7. Click **Save**.

Validate routing configuration

1. From a corporate **WhatsApp account**, send a test message to an external WhatsApp contact that triggers different DLP actions (Accept, Reject, Quarantine, Encrypt).
2. In the **DLP portal**, select **Incidents view** and confirm that the test message appears as a scanned list with X-header.
3. In the **WhatsApp gateway logs,** find the same message and verify that it includes the X-header with a DLP action.
    - If the header is **DLP-Accept**, confirm the external user received the message.

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- If the header is **DLP-Reject or DLP-Quarantine**, confirm the message was blocked as configured.

Using DLP X-header–based use cases

DLP for WhatsApp allows you to implement use cases that take different actions on messages based on the X-DLP-WhatsApp header value. Typical actions include blocking, quarantining, encrypting, dropping attachments, or redirecting messages to another workflow.

Before creating detailed use-case rules, ensure that:

- Outbound routing from the WhatsApp gateway to the DLP portal is configured and active.
- Inbound processing from the DLP portal back to the WhatsApp gateway is configured.

After this setup, you can define custom use cases in your WhatsApp gateway or policy engine according to your organization's data protection requirements.

Use case 1: Block card data to external contacts

Prevent users from sending WhatsApp messages that contain payment card data to external recipients.

Steps

1) In the **DLP portal**, go to **Policies > WhatsApp**.
   a) Click **Create policy** and name it **Block card data to external contacts**.
   b) In the **Conditions** section, add:
      i) **Content type**: Text and media.
      ii) **Data classifier**: Payment card data (PCI) or an equivalent built-in pattern.
      iii) **Direction/recipient**: Messages sent to external contacts.
   c) In the **Actions** section, set DLP response to **DLP-Reject**.
   NOTE: Ensure the policy is configured to add the X-DLP-WhatsApp header with value DLP-Reject when the rule matches.
   d) Set the policy priority **High** and click **Save**.
2) In the **WhatsApp admin console**, select **Inbound rules/DLP processing**.
   a) Edit the rule that inspects the **X-DLP-WhatsApp** header.
   b) Add a condition: If X-DLP-WhatsApp equals **DLP-Reject**.
   c) Set the action to:
      i) **Block the message**.
      ii) **Send a notification to the sender**.

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3) Validate the configuration

   a) From a test user, send a WhatsApp message with a dummy test card number to an external contact.

   b) In the **DLP portal**, confirm the incident is logged with the **DLP-Reject** action.

   c) In the **gateway logs**, verify that the message includes **X-DLP-WhatsApp: DLP-Reject** and that it is not delivered to the external user.

Use case 2: Quarantine confidential documents in external groups

Hold messages that contain confidential documents when sent to external WhatsApp groups, so security teams can review them before release.

Steps

1) In the **DLP portal**, go to **Policies > WhatsApp**.

   a) Click **Create policy** and name it **Quarantine confidential docs to external groups**.

   b) In the **Conditions** section, add:

      i) **Content type**: Files/media attachments.

      ii) **Data classifier**: Confidential label, internal-only tag, or document fingerprint.

      iii) **Direction/recipient**: Chats where at least one participant is external, and the conversation is a group.

   c) In the **Actions** section, set DLP response to **DLP- Quarantine**.
   NOTE: Ensure the policy is configured to add the X-DLP-WhatsApp header with value DLP- Quarantine when the rule matches.

   d) Set the policy priority **High** and click **Save**.

2) In the **WhatsApp admin console**, select **Inbound rules/DLP processing**.

   a) Edit the rule that inspects the **X-DLP-WhatsApp** header.

   b) Add a condition: If X-DLP-WhatsApp equals **DLP-Quarantine**.

   c) Set the action to:

      i) **Store the message in a Quarantine queue**.

      ii) **Send an alert to the security compliance channel**.

      iii) Optionally send a notification to the user that the message is under review

3) In the review console, the reviewers can:

   a) Open the quarantined item.

   b) Decide Approve (allow) or Deny (delete/keep quarantined).

4) Validate the configuration

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a) From a test user, send a WhatsApp message with a confidential file to an external WhatsApp group.

b) In the **DLP portal**, confirm the incident is logged with the **DLP- Quarantine** action.

c) In the **gateway logs**, verify that the message includes **X-DLP-WhatsApp: DLP-Quarantine** and that it is not delivered to the external user.

d) Group members do not receive the file.

e) The reviewer can see and act on the quarantined item.

Use case 3: Encrypt messages with personal data

Automatically encrypt WhatsApp messages containing personal data (PII) when they leave the organization, instead of blocking them.

Steps

1) In the **DLP portal**, go to **Policies > WhatsApp**.
   a) Click **Create policy** and name it **Encrypt messages with personal data**.
   b) In the **Conditions** section, add:
      i) **Content type**: PII classifiers
      ii) **Data classifier**: national IDs, phone numbers, customer IDs, email addresses.
      iii) **Direction/recipient**: Limit the scope to messages sent to external recipients.
   c) In the **Actions** section, set DLP response to **DLP- Encrypt**.
   d) NOTE: Ensure the policy is configured to add the X-DLP-WhatsApp header with value DLP- Encrypt when the rule matches.
   e) Set the policy priority **High** and click **Save**.
2) In the **WhatsApp admin console**, select **Inbound rules/DLP processing**.
   a) Edit the rule that inspects the **X-DLP-WhatsApp** header.
   b) Add a condition: If X-DLP-WhatsApp equals **DLP- Encrypt**.
   c) Set the action to:
      i) Replace the original message with a secure notification or
      ii) Use a built-in secure messaging/encryption mechanism.
      iii) Optionally send a notification to the sender.
3) In the review console, the reviewers can:
   a) Open the quarantined item.
   b) Decide Approve (allow) or Deny (delete/keep quarantined).
4) Validate the configuration
   a) From a test user, send a WhatsApp message containing PII to an external contact.

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b) In the **DLP portal**, confirm the incident is logged with the **DLP- Encrypt** action.

c) In the **gateway logs**, verify that the message includes **X-DLP-WhatsApp: DLP-Encrypt** and the content is received via the secure method.

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